Title: Phase 1 Report: Descriptive Analysis of Smoking and Elevated Blood Pressure
Course Project: From Statistical Foundations to Reproducible Research
Dataset: NHANES August 2021–August 2023
Date: 3 Nov 2026

1. Introduction and Statistical Question

This project investigates the association between cigarette smoking and elevated blood pressure among U.S. adults using data from the National Health and Nutrition Examination Survey (NHANES) 2021–2023 cycle.

The main statistical question addressed in this project is:
In U.S. adults aged 20 years and older, is the proportion of individuals with elevated blood pressure (systolic ≥ 130 mmHg or diastolic ≥ 80 mmHg) different between current cigarette smokers and non-smokers?

The unit of observation is the individual participant, identified uniquely by the variable SEQN The target population is the non-institutionalized U.S. adult population.

2. Data Files and Merging Process (Requirement A1)

To answer the research question, three data files from the NHANES 2021–2023 cycle were downloaded from the official CDC website and merged:

- DEMO L.xpt: Demographics (contains SEQN, age RIDAGEYR, sex RIAGENDER, and survey weights WTMEC2YR).
- BPXO L.xpt: Blood Pressure - Oscillometric Measurements (contains SEQN and blood pressure readings).
- SMQ L.xpt: Smoking - Cigarette Use (contains SEQN and smoking status variables).

The files were merged using the unique participant identifier SEQN via left joins, ensuring all participants from the demographics file were retained.

- **N in DEMO_L (initial):** 11,933
- **N in BPXO_L (exam):** 7,801
- **N in SMQ_L (questionnaire):** 9,015
- **N after merge:** 11,933 (11,933 unique SEQN values were retained, confirming no duplicate records were introduced during the merge).

3. Sample Weights and Complex Survey Design (Requirement A2)

NHANES uses a complex, multistage probability sampling design. Therefore, each individual in the sample does not represent just themselves, but rather a specific number of individuals in the target population. This is reflected in the sample weights.

Because our primary outcome variable (blood pressure) comes from the Mobile Examination Center (MEC) exam, the appropriate weight to use is the **MEC Exam Weight (**WTMEC2YR**)**. CDC guidelines state that when combining data from different components (e.g., questionnaire and examination), one should use the weight associated with the component with the lowest response rate (which is typically the MEC exam). Using this weight ensures that our estimates are representative of the entire U.S. non-institutionalized adult population, compensating for non-response and oversampling.

4. Analytic Population Definition (Requirement A4)

To define the analytic population, several exclusion criteria were applied sequentially. The flow of participants is documented below:

- **Start (after merge):** 11,933
- **Excluded due to age < 20 years:** 4,124 (Retained: 7,809)
- **Excluded due to missing or zero WTMEC2YR:** 1,745 (Retained: 6,064)
- **Excluded due to missing BP-HIGH or SMOKER status:** 208 (Retained: **5,856**)

The final analytic sample consists of **5,856** adults aged 20 years and older with valid exam weights and complete data for the variables of interest.

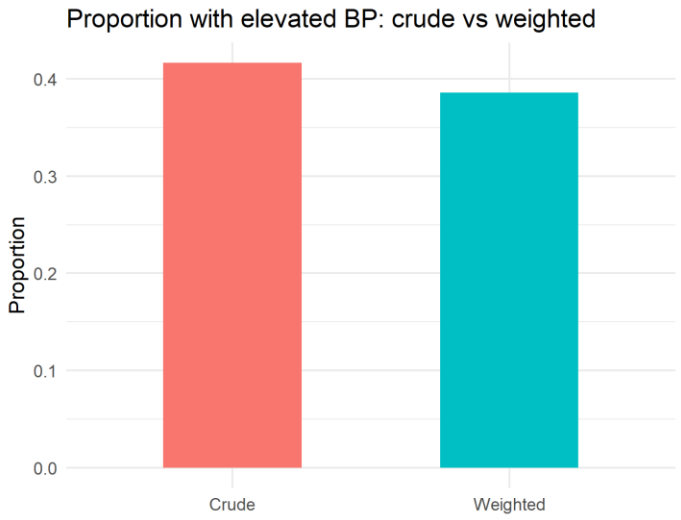
Variable Definitions:

- **Elevated Blood Pressure (BP-HIGH):** Coded as 1 if the mean of available systolic readings was ≥ 130 mmHg OR the mean of available diastolic readings was ≥ 80 mmHg; otherwise 0.
- **Smoking Status (SMOKER):** Coded as 1 if the participant reported smoking every day or some days (SMQ040 = 1 or 2); otherwise 0 (non-smoker).

5. Crude vs. Weighted Estimates (Requirement A3)

In this section, we compare the unweighted (crude) proportion of individuals with elevated blood pressure to the survey-weighted proportion.

- **Crude proportion:** 41.65%
- **Weighted proportion:** 38.57%



Discussion: The crude sample proportion (41.65%) overestimates the population proportion (38.57%). This difference occurs because the raw sample is not perfectly representative of the U.S. population. NHANES oversamples certain demographic groups (e.g., older adults, minority populations) who may have higher rates of elevated blood pressure. Applying the survey weights adjusts for this oversampling, providing a more accurate and unbiased estimate of the true population parameter.

6. Descriptive Comparison (Initial Comparison)

In Phase 1, we only provide a purely descriptive comparison of the proportion of elevated blood pressure between the two groups. No inferential claims or hypothesis testing are made at this stage.

The following table summarizes the crude and weighted proportions by smoking status:

Smoking Status	Sample Size (n)	Crude Proportion	Weighted Proportion
Non-Smoker	4,978	41.1%	38.1%
Current Smoker	878	44.9%	41.4%

Additionally, stratifying by sex reveals descriptive differences. For instance, the crude proportion of elevated blood pressure is higher in males (44.2% for non-smokers, 46.2% for smokers) compared to females (38.7% for non-smokers, 43.2% for smokers).



7. Conclusion of Phase 1

The data preparation and descriptive analysis for Phase 1 are successfully completed. The weighted estimates suggest that the proportion of elevated blood pressure is descriptively higher among current smokers (41.4%) compared to non-smokers (38.1%). However, these differences will be rigorously evaluated using confidence intervals and hypothesis testing in Phase 3.

For now, the analysis is descriptive, and all data preparation steps, including merging, weight selection, and handling of missing data, have been fully documented to ensure reproducibility.